



# Configurable DC-TO-DC CONVERTER MODULE Using Three ICs

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Most kits with embedded microcontrollers (MCUs) operate off a single power supply ( $V_{CC}$ ) source like 5V or 3.3V. But for some experiments you need one or more positive voltages higher than  $V_{CC}$  and one or more negative voltages lower than ground.

Often, the required current is in several milli-amperes (mAs). In such cases, you may use ICL7660, ICL7660A or similar integrated circuits (ICs) for the power supply. But voltage and current provided by a single ICL7660 is not sufficient, so you may need two or more ICs in parallel or cascaded to meet the requirement.

This article describes a universal configurable module that uses three ICL7660 ICs. The module is flexible and allows all basic configurations for ICL7660 and similar ICs.

## Circuit and working

ICL7660, MAX1044, ADM660, MAX660, MAX665, LTC1044, LTC1046 and some other similar ICs are DC-to-DC inductor-less converters. These ICs are not exact equivalents but have similar functions and are pin-to-pin compatible in many cases.

However, there are several important differences between them, including working frequency, maximal output current and power supply range. For example, LTC1046 can produce output current up to 50mA, ADM660 up to 100mA and ICL7660 up to 20mA.

The proposed module will work with the ICs listed above and similar ICs but its performance will depend on the IC used. The mentioned ICs are available in 8-pin PDIP packages. Short descriptions of their each pin are given in Table I.

TABLE I  
DESCRIPTION OF THE  
PINS OF ICL7660

| IC pin | Description                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Pin 1  | Not connected (NC) or boost pin                                                                                                           |
| Pin 2  | Pin for positive end of external capacitor                                                                                                |
| Pin 3  | For ground or for external offset voltage                                                                                                 |
| Pin 4  | Pin for negative end of external capacitor                                                                                                |
| Pin 5  | For negative output voltage                                                                                                               |
| Pin 6  | Low-voltage operation input. If power supply is below 3.5V, this input should be connected to ground. Threshold voltage depends on the IC |
| Pin 7  | Oscillator control input. This input has internal capacitor of around 15pF and can be driven from external oscillator                     |
| Pin 8  | For positive power supply. Maximum voltage difference between pin 8 (V+) and pin 3 (GND) depends on the ICs                               |

## PARTS LIST

|                                                               |                                |
|---------------------------------------------------------------|--------------------------------|
| <b>Semiconductors:</b>                                        |                                |
| IC1-IC3                                                       | - ICL7660 DC-to-DC converter   |
| D1-D9                                                         | - 1N5817 Schottky diode        |
| <b>Resistors (all 1/4-watt, <math>\pm 5\%</math> carbon):</b> |                                |
| R1-R3                                                         | - 100-ohm                      |
| <b>Capacitors:</b>                                            |                                |
| C1-C3                                                         | - 1 $\mu$ F, 25V electrolytic  |
| C4-C9                                                         | - 10 $\mu$ F, 25V electrolytic |
| C10-C18                                                       | - 47 $\mu$ F, 25V electrolytic |
| C19-C21                                                       | - 1nF ceramic disk             |
| <b>Miscellaneous:</b>                                         |                                |
| CON1, CON2                                                    | - 12-pin terminal connector    |
| J1-J21                                                        | - Shorting jumper              |
|                                                               | - 5V DC power supply           |

The ICs can work in several configurations. Input voltage can be inverted ( $V_{OUT} = -V_{IN}$ ), divided by 2 ( $V_{OUT} = V_{IN}/2$ ) or multiplied by  $n$  ( $V_{OUT} = \pm nV_{IN}$ ). The ICs can work with an internal or external oscillator.

Fig. 1 shows the circuit diagram of the configurable DC-to-DC converter using three ICL7660 ICs. The module has the following advantages:

1. IC1, IC2 and IC3 can be of same type or different.

2. Each IC can work with independent power supplies, namely, V1, V2 and V3.

3. Each IC can work in different configurations.

4. Outputs of the ICs can be paralleled for higher output current.

5. The ICs can be cascaded for higher voltages.

6. The ICs can work with external oscillators having the same or different frequencies.

7. Each module can produce single negative output voltage, approximately equal to  $-V1$ ,  $-V2$  and  $-V3$  at V4, V5 and V6, respectively. That is,

$$V4 = -V1$$

$$V5 = -V2$$

$$V6 = -V3$$

8. Each module can produce single positive output voltage at V7, V8 and V9, respectively, which is approximately equal to:

$$V7 = 2 \times V1 - 2 \times Vd$$

$$V8 = 2 \times V2 - 2 \times Vd$$

$$V9 = 2 \times V3 - 2 \times Vd$$

Where  $Vd$  is the voltage drop across the corresponding Schottky diodes in series.

9. The module can produce positive and negative output voltages. All diodes (D1 through D9) should preferably be fast-switching Schottky diodes like 1N5817/8/9. BAT series, like BAT42, BAT43, BAT47, BAT48, but other similar Schottky diodes can also be used. While choosing these diodes, forward voltage drop and reverse voltage of the diodes should be taken into consideration.

**Jumpers.** The ICs can be configured independently with several jumpers. Table II gives short descriptions of the functions of the jumpers